



How Does the Coupling of Real-World Policies with Optimization Models Expand the Practicality of Solutions in Reservoir Operation Problems?

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Abstract

This study first compares how different formulations of a reservoir operation problem with conflicting objectives affect the quality of the generated solution set. Six models were developed for comparative analysis: three using dynamic programming and three using the evolutionary multi-objective direct policy search (EMODPS) algorithm. Subsequently, to improve the quality of the generated solution set, an EMODPS model was selected and coupled with a zone-based hedging policy that has been currently applied in real-world reservoir operations. The proposed methodology was applied to a multipurpose reservoir in South Korea. Among the different models, the EMODPS-Gaussian model with three parameters outperformed dynamic programming models. Moreover, coupling the status quo zone-based hedging rule with the optimization model improved the average duration of failure from the supply side by 37.4%. On the other hand, when measured from the demand side, the results indicated that the magnitude of failure also improved by 4.15% at the cost of frequency of water deficit. The overall results of this study suggest that the integrative use of optimization models with hedging rules is potentially applicable in future drought mitigation measures.

Keywords Reservoir operation · Dynamic programming · Evolutionary multi-objective direct policy search · Zone-based hedging rule

1 Introduction

During the last half-century, the derivation of optimal reservoir control policies through the use of emerging technological advancements gained considerable popularity and has become a key topic in the field of water resources management (Yakowitz 1982; Yeh 1985; Klemesš 1987; Labadie 2004). Optimization techniques include linear programming (Belaineh et al. 1999), nonlinear programming (Birhanu et al. 2014), and dynamic

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programming (DP) (Stedinger et al. 1984). Thus far, these tools have been adopted as primary problem-solving tools for water resources engineers.

In particular, algorithms belonging to the DP family are prominent options applicable to the majority of real-world optimization problems. Given that the system dynamics of the target problem are well known (Giuliani et al. 2016a), DP transforms the convoluted original problem into a sequence of relatively simpler problems. By approximating the state and decision space by discretization, DP provides the optimal set of decision variables corresponding to every possible combination of state variables. Because this procedure can be applied to all types of objective functions or constraints regardless of their shapes, DP and its variations (Stedinger et al. 1984; Saadat and Asghari 2019) have been continuously applied to numerous reservoir optimization problems.

Despite the wide applicability of DP, approximation procedures that are compulsory in the DP framework may cause decision biases that potentially limit optimal decision making (Kasprzyk et al. 2009). Thus, efforts have been extended to foster optimization techniques to derive more precise resolutions to multi-objective problems involving conflicting trade-offs (Reed et al. 2013). One of the main approaches in this context is the adaptation of multi-objective evolutionary algorithms (MOEAs) (Javadi et al. 2015; Liu et al. 2019a, b), in which the algorithm aims to address the target problem by making the least possible simplifications. Specifically, the evolutionary multi-objective direct policy search (EMODPS) framework (Giuliani et al. 2016a), which combines direct policy search (DPS) (Rosenstein and Barto 2001), nonlinear approximating networks, and MOEAs into a simulation-based optimization algorithm, has been proven to be more effective in yielding solutions to certain problems (Salazar et al. 2017; Quinn et al. 2019). Because solutions generated from EMODPS may yield more flexible decisions under more varied circumstances, EMODPS has been employed in complex multi-objective applications in which obtaining reference optimal solutions is relatively demanding. As a reference, Giuliani et al. (2016a) demonstrated that the use of EMODPS with both artificial neural networks (ANNs) and radial basis functions (RBFs) and proved that EMODPS model with RBF outperformed the ANN models in terms of both 75% and 95% thresholds.

Although operating policies derived via optimization schemes such as DP and EMODPS may theoretically guarantee optimal alternatives, several factors have hindered their implementation in real-world reservoir operations (Quinn et al. 2019), including the extremely risk-averse nature of decision makers, administrative or legal constraints, and the lack of expert knowledge among related stakeholders. Alternatively, decision makers have adopted various types of hedging rules on-site, such as two-point (Bayazit and Ünal 1990), continuous (Hashimoto et al. 1982), and zone-based hedging (Shih and ReVelle 1995), which conserve a portion of reservoir storage in case of future droughts (You and Cai 2008a, b). In particular, zone-based (discrete) hedging rules are popular in on-site reservoir operations and have been commonly adopted owing to their simple methodology and relative ease of implementation for dam operators with limited advanced expert knowledge. The zone-based hedging rule first prioritizes water according to its uses and sequentially enforces restrictions in the case of droughts. Consequently, these earlier restrictions would prevent a reservoir from suffering from severe water shortages in the future.

Owing to the discrepancy between the solutions derived from optimization algorithms and hedging rules, only a few studies have endeavored to merge the concepts of policies derived from optimization algorithms and policies being implemented onsite. As an extension of participatory or collaborative decision-making efforts in water resources management, this study intends to link theoretical optimization and real operation through the following procedures: (1) generate optimal solution sets of a multi-objective reservoir

operation problem solely using distinctive optimization algorithms based on K-fold cross-validation and compare the quality of the derived solutions, (2) compare their performances with those of actual operations executed based on status quo (SQ) rules, and (3) implement real-world hedging rules to the solutions from the optimization algorithms and explore the trade-off relationship between conflicting stakeholders based on risk, resiliency, and vulnerability. The authors intend to contribute both to the algorithmic component and to real-world drought response measures by providing a thorough analysis of multiple optimization frameworks and relating them to real-world operating policies.

2 Theoretical Methodologies

2.1 Dynamic Programming

DP is an optimization algorithm that searches for an optimal set of decision variables that maximizes the current and the expected benefits from future operations (Tejada-Guibert et al. 1995), and its variations have been applied to multiple water resources management problems (Yakowitz 1982; Yang et al. 2009; Fayaed et al. 2013; Feng et al. 2020). For the calculations, the DP first discretizes the entire range of states, decisions, and disturbance variables with appropriate intervals. Then, the objective functions and constraints are iteratively calculated in discrete time steps across the entire optimization time horizon. Consequently, a steady-state solution or an optimal set of decision variables corresponding to each value of the state variables is computed as the outcome of the optimization procedures.

For example, let us consider a multi-objective problem in which the aim is to maximize a series of objectives, $\mathbf{J} = J^1, J^2, \dots, J^k$ by deciding on the optimal policy of release decisions, R_t^* across the optimization horizon T . Assuming perfect knowledge of the future inflow Q_t , this optimization problem can be solved using a deterministic DP (DDP) model by solving the following functional equation (i.e., recursive equation) under the time horizon:

$$f_{opt}^n(S_t, Q_t) = \max_{R_t^*} \left[Z_t(S_t, Q_t, R_t) + \gamma f_{opt}^{n-1}(S_{t+1}, Q_t) \right] \quad (1)$$

subject to

$$S_{t+1} = S_t + Q_t - R_t \quad (2)$$

$$S_{min} \leq S_t \leq S_{max} \quad (3)$$

$$R_{min} \leq R_t \leq R_{max} \quad (4)$$

where $t = 1, \dots, T$; T = final time step; n = the number of time periods remaining until the end of the optimization horizon; f_t^n = the value of the benefit function from the current period to the end of time horizon T ; Z_t = the value of the aggregated objective function during time period t with prescribed weights (i.e., $Z_t = \sum_{i=1}^k w_i \times J_t^i$); S_t = the storage at the beginning of period t ; Q_t = the inflow during period t ; R_t = the actual release during period t ; γ = a discount factor; S_{min}, S_{max} = the minimum and maximum storage, respectively; and R_{min}, R_{max} = the minimum and maximum release, respectively.

Because this specific formulation of the problem includes two state variables (S_t and Q_t), and a single decision variable (R_t), solving Eq. 1 yields an optimal set of decision variables, R_t^* with dimensions proportional to the discretization intervals of the state variables and the length of the optimization horizon. Because DP manages both nonlinear and multi-objective characteristics of complex reservoir operational design problems (Giuliani et al. 2016a) in its objective function, it has been considered one of the most prominent options for solving water resources management problems. In particular, the stochastic version of DP (SDP), which addresses future uncertainties with hydrological state variables or transition probabilities, has been adopted in multiple reservoir operation problems (Stedinger et al. 1984; Tejada-Guibert et al. 1995; Kim et al. 2021).

Despite its advantages, few scholars have argued for the application of DP to more complicated water resources problems because of its notorious curses: curse of dimensionality (Bellman 1957), curse of modeling (Tsitsiklis and Van Roy 1996), and curse of multiple objectives (Powell 2007). Briefly, the curse of dimensionality refers to the exponential increase in the number of computations proportional to the number of state variables, the curse of modeling refers to the necessity of including all input variables in the model as supplementary state variables, and the curse of multiple objectives refers to the exponential increase in the number of computations proportional to the number of objectives. This has motivated recent technical advancements aimed primarily at overcoming these three curses by using novel approaches other than DP (Bertsekas 2005; Giuliani et al. 2016a).

2.2 Evolutionary Multi-Objective Direct Policy Search

EMODPS is an example of the parameterization-simulation-optimization (PSO) approach (Koutsoyiannis and Economou 2003), where the MOEAs are merged with DPS and nonlinear approximating networks to derive a set of optimal policies. Instead of implementing approximation during problem formulation processes, such as discretization in DP, EMODPS directly optimizes the parameters of the optimal policy through a simulation-based optimization scheme. It is argued that the unique features of EMODPS partially resolve the three curses of DP: (1) the curse of dimensionality by evading the iterative computation of the value of the benefit function; (2) the curse of modeling by using a simulation model that can include exogenous information directly during simulation; and (3) the curse of multiple objectives by attaining the entire Pareto front through the adoption of MOEAs during its search (Kasprzyk et al. 2009; Giuliani et al. 2016a).

The problem formulation and optimization conducted through the EMODPS framework can also be illustrated through the same problem that maximizes a series of objectives, $\mathbf{J}$. The EMODPS aims to search for a set of parameters ($\Theta^* = \Theta_1^*, \dots, \Theta_n^*$) that constitute the optimal policy p_θ^* . The formulation of the optimal policy that maximizes $\mathbf{J}$ can be expressed by the following equation Eq. 5:

$$p_\theta^* = \arg \min_{p_\theta} \mathbf{J}_{p_\theta} \quad (5)$$

The policy parameter vector Θ that constitutes the operating policy p_θ through parameterization, is coupled with diverse types of networks of functions that are flexible in shape. Typical examples of network functions include ANNs and RBFs (Park and Sandberg 1991). The following example demonstrates the RBF with cubical shape, where the decision variable R_t is decided according to both the shape of the functions and the values of the parameters Θ (Giuliani et al. 2016a):

$$R_t = \min \left(\max \left(\sum_{j=1}^n \theta_{j,1} \left| \frac{S_t - \theta_{j,2}}{\theta_{j,3}} \right|^3, R_{\min} \right), R_{\max} \right) \forall t, i \quad (6)$$

where n = the number of RBFs; $\theta_{j,1}, \theta_{j,2}, \theta_{j,3}$ = a set of parameters that compose the j th RBF; and S_t = the value of the state variable at time t (storage at time t in this example).

The last component in the EMODPS framework, MOEA, is adopted during the search procedures of the optimal policy parameter set Θ^* from Eq. 6. Because MOEAs can discover the critical trade-offs facing water resources systems (Reed et al. 2013), non-dominated or Pareto optimal sets of solutions can be derived without aggregating multiple objectives into a single function. Consequently, policymakers are presented with a full tradeoff and can express their preferences for a single or several alternatives among the non-dominated solutions. To generate a more precise set of solutions, improvements in diverse MOEA schemes have been developed, including the niched Pareto genetic algorithm (Horn et al. 1994), and the non-dominated sorting genetic algorithm II (NSGA-II) (Deb et al. 2002). For detailed description on the performances of different MOEAs, refer to the papers by Nicklow et al. (2010), Hadka and Reed (2012), and Reed et al. (2013).

2.3 Zone-based Hedging Rules

Operating reservoirs with alternatives derived from optimization algorithms, such as DP or EMODPS, would theoretically lead to higher measures of superior performance with respect to optimized objectives. However, the on-site application of these policies may be limited by the lack of theoretical backgrounds among decision makers (Seo et al. 2019), both actual and perceived errors in the models/forecasts, and the problematic attributes of systems models that are difficult to explain or understand (Quinn et al. 2019). Instead, the adaptation of relatively simpler policies such as diverse variants of hedging rules, or drought contingency plans, is more prevalent in real-world reservoir operations. Among various alternatives, zone-based (discrete) hedging rules (Shih and ReVelle 1995), in which the release decisions are determined according to the storage at a specific time period, are considered to be primarily due to their relatively simple and intuitive application procedures.

An explanation of zone-based hedging rule adopted by all 20 multipurpose dams in South Korea as their main operating policy is provided as follows. Operating a reservoir solely based on the zone-based hedging rule requires the following procedures: first, the reservoir's total demand is prioritized according to its use. In the case of South Korea, water use from a multipurpose dam is mainly classified into municipal (M), industrial (I), agricultural (A), and environmental (E), and priorities are granted to each category accordingly. For instance, municipal and industrial water uses have the highest priorities, followed by irrigation, and environmental uses. Finally, the release R_t is determined according to the storage zone, and can be expressed as Eqs. 7–11, which are considered the key operating procedures for the zone-based hedging rules:

$$R_t = D_t, \text{ when } S_{t-1} \in S_{t, \text{Normal Zone}} \quad (7)$$

$$R_t = \alpha_t \times D_t, \text{ when } S_{t-1} \in S_{t, \text{Zone 1}} \quad (8)$$

$$R_t = \beta_t \times D_t, \text{ when } S_{t-1} \in S_{t, \text{Zone 2}} \quad (9)$$

$$R_t = \gamma_t \times D_t, \text{ when } S_{t-1} \in S_{t, \text{Zone } 3} \quad (10)$$

$$R_t = 0, \text{ when } S_{t-1} \leq S_{DSL} \quad (11)$$

where α_t , β_t , and γ_t ($1 > \alpha_t > \beta_t > \gamma_t > 0$) = weighting factors determined according to the reservoir and season; D_t = the aggregate demand during month t ; S_{DSL} = the dead level storage of the reservoir.

This study aims to use of the current zone-based SQ policy in two manners: (1) compare the performance of the solutions derived from optimization algorithms with the actual performance based on the SQ policy and (2) couple the SQ policy during the simulation step of the PSO algorithm to analyze the effect of the incorporation of the hedging rule into an optimization algorithm.

3 Model Development for Boryeong Multipurpose Dam

3.1 Study Site

In this study, the Boryeong Dam, a multipurpose reservoir located on the western coast of the Korean Peninsula, was selected as the test case, and the proposed methodology to derive optimal release sequences with multiple alternatives was applied. Boryeong Dam supplies water for M&I, A and E uses to eight nearby administrative districts: Taean, Seosan, Dangjin, Yesan, Hongseong, Boryeong, Cheongyang, and Seochon (Fig. 1). Because of the marginal difference between the capacity of the reservoir ($116.9 \times 10^6 m^3$) and the aggregate annual demand of districts ($106.6 \times 10^6 m^3$), management of the reservoir during the drawdown period is even more critical. Moreover, differently from other typical reservoirs, the majority of the water demand from the Boryeong Dam is concentrated on M&I uses (83%). This high ratio for M&I water demand consequently increases the extent of the potential exposure of the public to drought damage if stocks are mismanaged.

To demonstrate how different optimization algorithms or problem formulation methodologies may ameliorate the performance of the reservoir, this study derived the optimal monthly release sequences for 20 historical water years (October 1999 through September 2019) using K-fold cross-validation. In this study, the observed inflow series were grouped into four subgroups ($K=4$) each with 15 years of inflow series for optimization, and the remaining five years of inflow for validation. During validation, the starting storage values were set as the actual storage values recorded at the beginning of the validation period (Table 1). The performance of each model was then comparatively analyzed. The overall flow of this study is shown in Fig. 2.

3.2 Optimization and Reservoir Simulation

This study adopted the multi-objective optimal operations (M3O) toolbox developed by Giuliani et al. (2016b) to derive the optimal release rules with different alternatives. In total, six different models were used to derive six different optimal release rules for each subgroup: the DDP model with perfect forecast (DDP-Perfect), DDP model with average inflow forecast (DDP-Average), a stochastic DP model assuming a lognormal distribution of the inflow (SDP), and three EMODPS models with different approximating

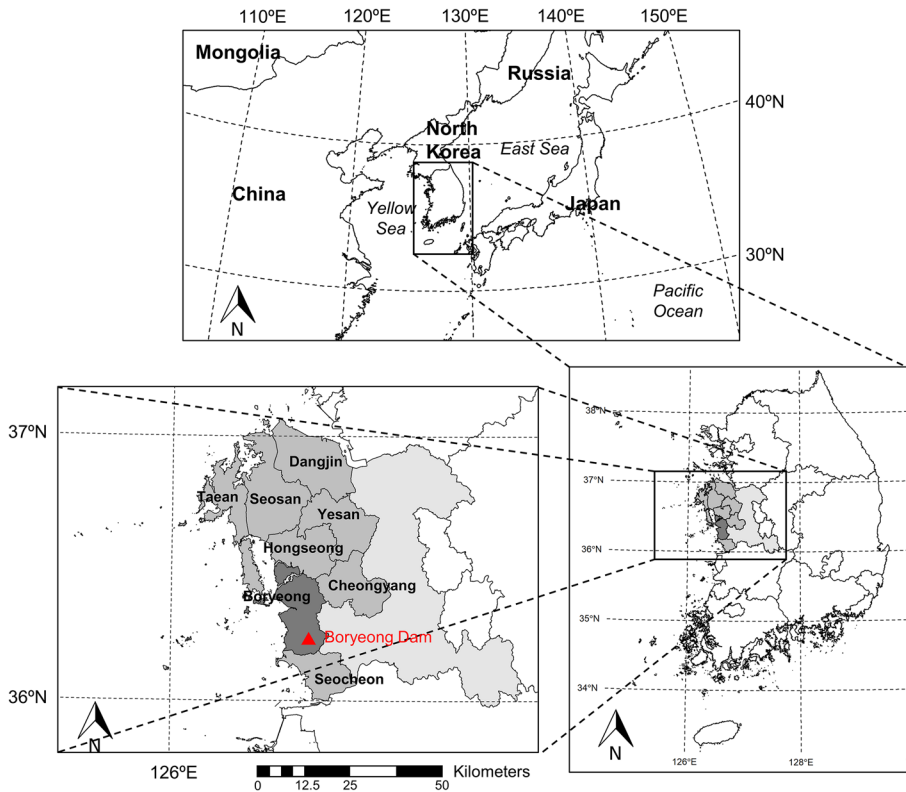


Fig. 1 Map of Boryeong Dam and the eight administrative districts that receive water from the reservoir

functions each with a distinct number of parameters (EMODPS-Linear, EMODPS-Root, and EMODPS-Gaussian).

In general, all models are operated mainly through the following mass balance equation Eq. 12:

$$S_{t+1} = S_t + Q_t - R_{n,t}^* \quad (12)$$

where S_t = the storage at the beginning of month t , Q_t = the inflow during month t , and $R_{n,t}^*$ = the optimal release decision during month t derived through the n th model ($n = 1, 2, \dots, 6$).

All models attempt to minimize the two objectives that reflect the preferences of two conflicting groups of stakeholders: J^{supply} , which reflects the interests of reservoir managers and J^{demand} , which reflects the satisfaction of water users. Both objectives are measured in terms of magnitude and expressed as the average percentage of deficit over the entire optimization (or simulation) time horizon H . The first objective minimizes the deficit between the Zone 2 storage and the actual storage in each time step, and is expressed as a percentage:

Table 1 Specific formulations for each subgroup for K-fold cross validation

K	Optimization Period (15 Years)	Validation Period (5 Years)	Average Inflow during Validation Period (m^3/sec)	Starting Storage (Million cubic meters, date)
1	October 2004 – September 2019	October 1999 – September 2004	4.64	99.597 (Sep-30, 1999)
2	October 1999 – September 2004, October 2009 – September 2019	October 2004 – September 2009	3.65	89.978 (Sep-30, 2004)
3	October 1999 – September 2009, October 2014 – September 2019	October 2009 – September 2014	4.39	64.365 (Sep-30, 2009)
4	October 1999 – September 2014	October 2014 – September 2019	2.72	47.364 (Sep-30, 2014)

$$J^{supply} = \frac{1}{H} \sum_{t=1}^H \max \left(\frac{S_t^{target} - S_t}{S_t^{target}}, 0 \right) \times 100\% \quad (13)$$

where S_t = the actual storage at the beginning of month t , and S_t^{target} = the predetermined Zone 2 storage during month t .

The second objective is to minimize the average deficits between releases and demands, which are expressed as a percentage:

$$J^{demand} = \frac{1}{H} \sum_{t=1}^H \max \left(\frac{D_t - R_t}{D_t}, 0 \right) \times 100\% \quad (14)$$

where D_t = the monthly aggregate demand during month t , and R_t = the actual release during month t . Ideally, both J^{supply} and J^{demand} should be zero, which indicates that the actual storage is at least above the target storage during the entire simulation period, and the release is higher than the demand. The two contrasting objectives in this study, J^{supply} and J^{demand} , were selected to reflect the real-world conflicts that have historically have occurred (Kim et al. 2019a, b).

3.2.1 Policy Derived using DP

In this study, three models were developed using DP as their primary optimization algorithm with different forecast information: DDP-Perfect, DDP-Average, and SDP. Primarily, the DDP-Perfect model assumes perfect forecasting during the entire optimization horizon of the future inflow. The solutions derived by the DDP-Perfect model were the reference set, in which the approximated solutions were not able to predominate. Conversely, the DDP-Average model was set to make release decisions that assumed the average inflow throughout the optimization horizon. Finally, the SDP model developed in this study assumes a lognormal distribution of the inflow series and derives the optimal release decisions based on Eq. 15:

$$f_{opt}^n(S_t, Q_t) = \max_{R_t^*} \left[Z_t(S_t, Q_t, R_t) + E_{Q_{t+1}|Q_t} f_{opt}^{n-1}(S_{t+1}, Q_{t+1}) \right] \quad (15)$$

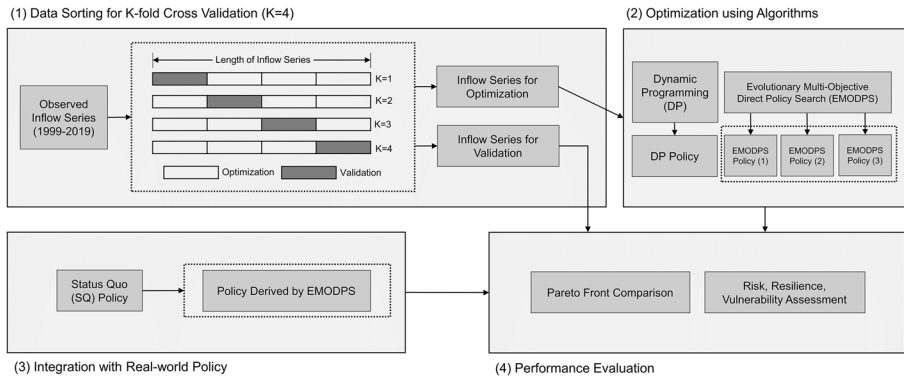


Fig. 2 Research flow chart adopted in this study

Because all DP-based models require a commensurate objective function, $Z_t(\cdot)$, for calculation in problems with two or more objectives, the aggregation was conducted as shown in Eq. 16 with combinations of weighting factors w_1 and w_2 :

$$Z_t = w_1 \times Z_{1,t} + w_2 \times Z_{2,t} \quad (16)$$

where w_1 and w_2 = weights multiplied by each of the objectives ($0 < w_1, w_2 < 1$); $Z_{1,t}$ and $Z_{2,t} = J^{supply}$ and J^{demand} calculated at month t , respectively.

3.2.2 Policy Derived using EMODPS

Because the type of selected optimization algorithm and the formulation of the approximation function affect the overall quality of the generated solution sets, this study generated manifold alternatives with EMODPS by varying the shape and number of parameters constituting the RBFs. Because it has been proven that the solution sets are sensitive to the shape of the parameterized function by Giuliani et al. (2016a), three shapes of RBFs with different numbers of parameters were adopted in this study: a 4-parameter linear RBF (EMODPS-Linear), a 6-parameter root-shaped RBF (EMODPS-Root), and a 3-parameter Gaussian RBF (EMODPS-Gaussian). All formulations adopted the NSGA-II for optimization because of its extensive use in water resources optimization problems and for equitable comparison.

The first alternative, the EMODPS-Linear model, was constructed using linear functions with four policy parameters (m_1, m_2, h_1 , and h_2). Consequently, the optimal release decision derived via the linear RBF was determined according to Eqs. 17–19:

$$L_1 = D_t + \tan(m_1) * (S_t - h_1) \quad (17)$$

$$L_2 = D_t + \tan(m_2) * (S_t - h_2) \quad (18)$$

$$R_t = \max(\min(L_1, D_t), L_2) \quad (19)$$

The second model adopted the Gaussian RBF, which is one of the most prominently adopted shapes of RBF in the field of scientific and engineering computing to determine

the policy for optimal release sequences. The three policy parameters that constitute the simplified Gaussian RBF (w_1, c_1 , and b_1) determine the decision R_t using the following equation:

$$R_t = \min \left(\max \left(\sum_{i=1}^j \left(D_t - \exp \left(w_i * \frac{S_t - c_i}{b_i} \right) \right), R_{\min} \right), R_{\max} \right) \quad (20)$$

The final shape of the RBF used in this study was a root-shaped RBF with six policy parameters (w_1, w_2, r_1, r_2, d_1 and d_2), and the release decision was derived according to the following equation Eq. 21:

$$R_t = \min \left(\max \left(\sum_{i=1}^j \left(w_i * \left| D_t - \frac{S_t - d_i}{r_i} \right|^{0.5} \right), R_{\min} \right), R_{\max} \right) \quad (21)$$

It has been proven that both the population diversity and convergence of solutions to the Pareto optimal front are paramount in several studies that apply MOEAs. Thus, to ensure both diversity and convergence of optimal solution sets, all models in this study adopted 40 individuals each with 500 generations during optimization. The generated optimal solution sets were then analyzed to identify more effective and reliable operating schemes during drought periods.

3.2.3 Incorporation of SQ Policy during Simulation

To analyze the effect of incorporating the hedging rule, this study coupled the real-world SQ policy during simulation in drought periods with the release decisions from the optimization algorithm. Among the types of hedging rules, zone-based hedging is used to operate the Boryeong dam. Under this rule, the actual release R_t is determined according to the zone to which storage S_t belongs. In this study, the SQ operating policy was included in the simulation step according to Eqs. 7–11, and real-world monthly rationing factors were adopted for a realistic simulation. Finally, the efficacy of incorporating hedging rules was analyzed in terms of frequency, duration, and magnitude from the stances of both water suppliers and users.

4 Results and Discussion

4.1 Comparison in Performance of Models

The application of the proposed methodology at the reservoir on the study site resulted in diverse sets of solutions, each distinct in quality and diversity. For comparative analysis of their performance, the solutions generated by the DDP-Perfect, DDP-Average, SDP, EMODPS-Linear, EMODPS-Root, and EMODPS-Gaussian models were plotted and compared with the actual operating results of the same period, which include on-site, real-time decision making by experts in reservoir operations (Fig. 3).

Multiple insights can be derived according to the Pareto fronts generated using different models. Primarily, solutions derived by the optimization algorithms outperformed the results from the actual operation, as indicated by the red diamonds in Fig. 3, in three of the four subgroups. Although on-site release decisions were made with more information than those in the optimization algorithms, the results indicated that the adaptation of any type

of optimization model to actual reservoir operation would lead to improved performance under normal conditions. However, the actual operation resulted in a better performance in the case of droughts (Fig. 3d). This result indicates that the SQ policy, on which the actual operations are based, may yield more practical outcomes when adequately employed under suitable conditions.

Among the optimization models, the DDP-Perfect model, which assumed perfect forecasting of future inflow series, yielded the best performance, as initially expected in all subgroups. Of the three EMODPS models, the EMODPS-Gaussian model with three parameters yielded the results closest to the ideal point among all subgroups, represented as a blue star in Fig. 3. Policies derived by the EMODPS-Linear model followed those derived from the EMODPS-Gaussian model regarding proximity to the reference set; however, they outperformed the other two models regarding solution diversity. Despite having more parameters than the other two, the EMODPS-Root model performed worst in terms of proximity to the ideal point. This indicates that the quality of the solutions generated by the EMODPS algorithm was more sensitive to the shape of the network functions than the number of parameters that constitute them. Moreover, in cases where deriving both optimal and diverse solution sets is challenging, the overall trade-off relationship between the performance and the number of generated solutions, which can be explored from the results of the EMODPS-Linear and EMODPS-Root models, stresses the importance of the decision makers choosing optimal operating policies depending on their circumstances.

Finally, from Fig. 3, the alternatives derived from the DDP-Perfect model may be considered as the optimal set under which, in all circumstances, the reservoir should be operated because their performance is adjacent to the ideal point. However, deterministic models are types of open-loop control, where the sequence of actions is not updated as a function of new observations of state variables (Herman et al. 2020). Because of this attribute, the optimal release decisions made under the deterministic models would severely underperform in the case of unexpected states. This can be proved by the inferior performance of the DDP-Average model in two of the subgroups, $K=2$ and $K=4$ (Fig. 3b, d, respectively).

Conversely, the SDP model and all types of EMODPS models, which belong to the group of policy-searching closed-loop models, update their release decisions dynamically according to either the present or future states. This indicates that the closed-loop models would be more flexible to future uncertainties and would surpass the performance of deterministic models in cases of unanticipated incidents such as floods or droughts. Moreover, the release decisions from the EMODPS models are defined under the entire domain of state variables, whereas the decisions from the DP models are only defined in limited domains because of the discretization of the state variables. Thus, policies from adaptive optimization algorithms with approximating functions of various shapes should be considered when deriving optimal solutions for reservoir operation problems.

4.2 Effect of Coupling Zone-based Hedging Rules During Droughts

In this section, the effect of coupling the zone-based hedging SQ policy with optimal solutions from the optimization algorithm during droughts was investigated. Given the total precipitation during the optimization horizon and the conflicting nature of the objectives, the relocation of the Pareto optimal solution sets closer to the ideal point is hardly achievable. Hence, this portion of the study explored the expansion of the trade-off relationship between water users and suppliers that may arise from the coupling zone-based hedging

policy during the simulation. Risk, resiliency, and vulnerability (Hashimoto et al. 1982) were chosen as performance indices to measure the frequency, duration, and magnitude of predefined failures. Higher values of risk, resiliency, and vulnerability correspond respectively to more frequent, longer-lasting, and higher failures, respectively.

To quantify the effect of the hedging rule under drought conditions, this study selected the EMODPS-Gaussian model under subgroup K=4, which yielded both optimal and diverse solution sets. The same set of derived solutions was simulated twice under the historical inflow, once based on the real-world hedging rule and once without it. Supply side failures were defined as those not meeting the target storage (i.e., $J^{supply} > 0$). Conversely, failure from the demand side was split into different uses and defined as when the release was less than the target demand. Because in the ratio for M&I uses, A uses are dominant (89% in the study reservoir) the risks and vulnerabilities for M&I and A uses were explored.

Figure 4 shows the performance indices from the supply side. Because the zone-based hedging rules restrict release from the reservoir in the case of relatively lower storage values, the average performance of the system measured from the supply side improved by 25.02% in risk, 37.38% in resiliency, and 24.23% in vulnerability. In particular, the maximum duration of failure, measured in terms of resiliency, was significantly reduced by 45.59% after incorporating hedging rules (Fig. 4b) from 17 months to 9.25 months. This indicates that the adaptation of hedging rules may potentially lead to less frequent, shorter, and smaller failures in case of droughts. Figure 5 depicts the comparative performances before and after coupling hedging that was calculated from the demand side. Owing to the conflicting nature of supply and demand, the frequency of failure measured from the demand side increased after hedging for both A and M&I uses (Fig. 5a, c). However, the average magnitude of failure slightly improved in the stances of water users (1.68% for A and 6.68% for M&I)

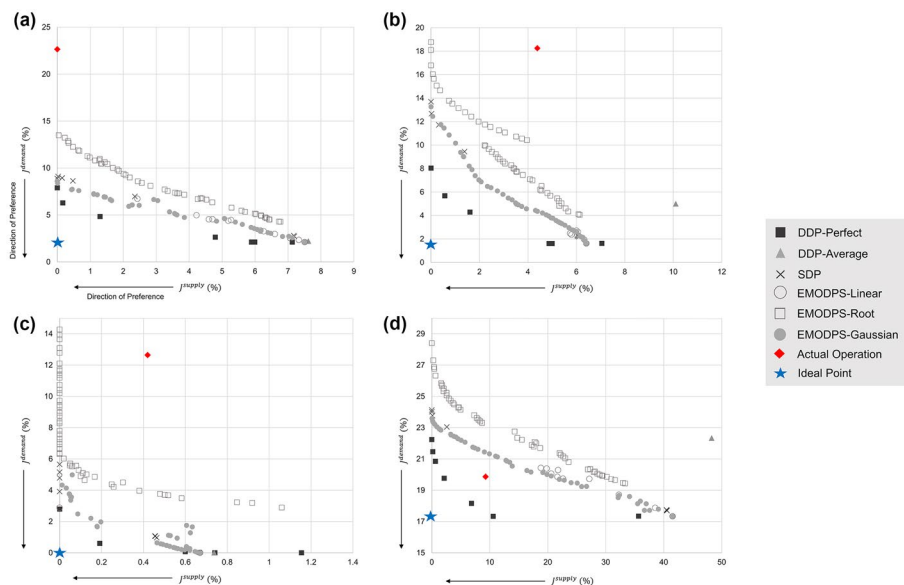


Fig. 3 Pareto fronts of the solution sets derived by each subgroup in K-fold cross-validation: (a) subgroup K=1, (b) subgroup K=2 (c) subgroup K=3, and (d) subgroup K=4

(Fig. 5b, d). This result indicates that when hedging is applied, the users experience more frequent failures, but with a downsized magnitude of water deficit. Considering the extremely risk-averse nature of water users facing water deficiency, reducing the magnitude of failure by incorporating hedging seems to be a better strategy given the increased risk of droughts in the future. Overall, these results indicate that the coupling of theoretical algorithms with real-world policies enlarges the range of choice with respect to vulnerability, although the previous studies primarily focused on the advancement of optimization techniques.

5 Conclusions

The primary motivation for this study was to conduct a comparative analysis between distinct formulations of the optimization algorithms used in reservoir operation problems and with actual operation results. Consequently, six different models were formulated using two eminent optimization algorithms: DP and EMODPS. The performance of all optimization models, measured as the magnitude of the deficit compared to the target storage and demand, surpassed the results from the actual operation with real-time calculations and experience of related experts under normal inflow conditions. Among the optimization models, solutions from the EMODPS models dominated those from the DP models in their proximity to the ideal point and variety of solutions across the entire Pareto front.

Another objective of this study was to combine the real-world zone-based hedging rule into the simulation process of optimally derived policy using an optimization algorithm, in which few studies have succeeded thus far. The solutions from the EMODPS-Gaussian model were simulated under a historical inflow series with and without coupling the real-world hedging policy. Despite the expected improvement in supply indices, the magnitude of failure from the demand side also increased after the incorporation of real-world policy. Considering the highly risk-averse nature of water users, the decrease in the average size of damage at the expense of increased frequency may be worthwhile in actual operations.

Ultimately, the findings from this study is not limited to the Boryeong Dam and can be extended to other reservoirs that are highly vulnerable to droughts with conflicting objectives, and rely on relatively simple hedging rules. The analytical framework of this study may also be expanded to include complex reservoir network systems with a cascade of

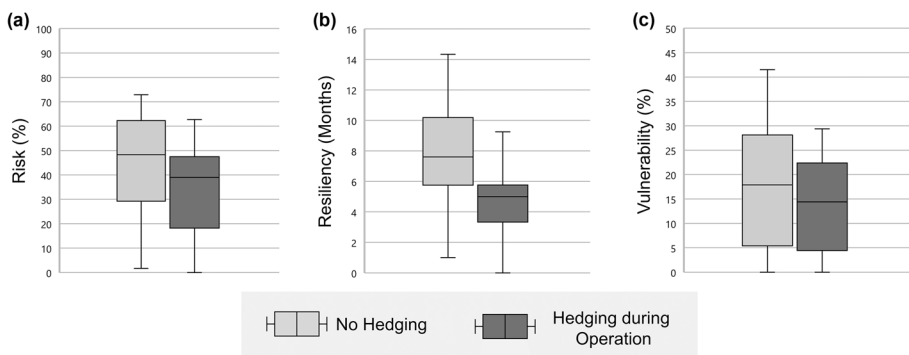


Fig. 4 Performance indices measured from the supply side: (a) risk, (b) resiliency, and (c) vulnerability

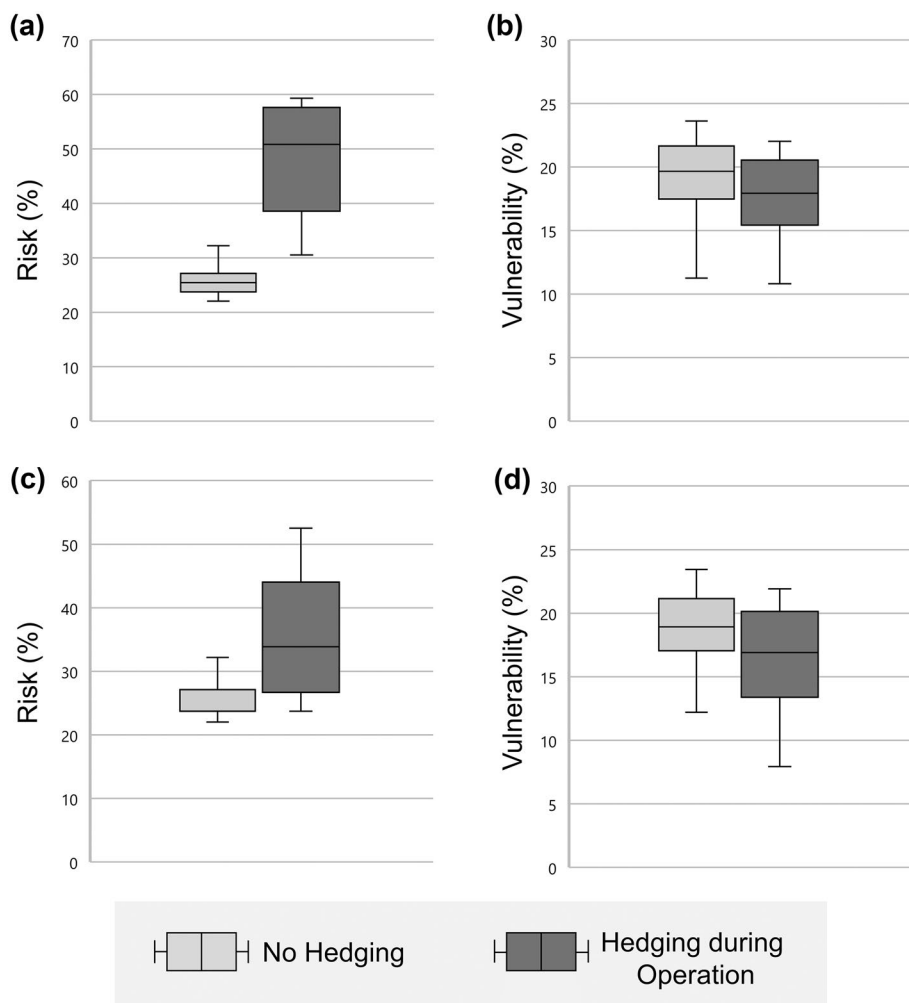


Fig. 5 Performance indices measured from the demand side: **(a)** risk of water for A uses, **(b)** vulnerability of water for A uses, **(c)** risk of water for M&I uses, and **(d)** vulnerability of water for M&I uses

reservoirs with different objectives, such as water-energy nexus (Barros et al. 2003; Liu et al. 2019a, b). Despite multiple findings from the application of the proposed methodology, improvements can be achieved in future studies. In particular, the values of different MOEAs could be explored by adopting various types of MOEAs during the optimization process. Conversely, the simulation under future inflow scenarios that reflect the increased risk of future climate variability, particularly from climate change, and analysis of the proposed alternatives under a wider climate range would enable the assessment of their performance under even more extreme climate conditions.

Authors' Contributions Gi Joo Kim and Young-Oh Kim contributed to the study conception and design. Material preparation, data collection and analysis were performed by Gi Joo Kim. The first draft of the

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Availability of Data and Material The datasets generated and analyzed during the current study are available on Mendeley Data (<https://data.mendeley.com/datasets/9ntcdscwpd/4>).

Code Availability All codes that support the findings of this study are available on Github (https://github.com/gijoo-kim/M3O_Boryeong_application).

Declarations

Ethics Approval Not applicable, because this article does not contain any studies with human or animal subjects.

Consent to Participate Not applicable, because this article does not contain any studies with human or animal subjects.

Competing Interests The authors have no conflicts of interest to declare that are relevant to the content of this article.

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